What is Green IT:

1.Green IT means using information technology in an environmentally friendly way to reduce harm to the environment while improving social and organizational performance.

2.It focuses on saving energy, reducing pollution, lowering carbon emissions, conserving natural resources, creating green jobs, and promoting digital inclusion to support sustainable development.

Why GT is Needed?

1.Modern development is heavily dependent on fossil fuels and mass production, which leads to several environmental problems such as:

- Climate change and global warming
- Water and soil pollution
- Depletion of natural resources
- Loss of biodiversity

2.Green Technology (GT) emerges as a solution to these environmental challenges by providing cleaner and more efficient alternatives to traditional technologies.

3.GT helps in reducing pollution, conserving natural resources, and promoting sustainable development for future generations.

4.It also encourages the use of renewable energy sources such as solar, wind, and hydro power, which are environmentally friendly.

5.Green Technology supports energy efficiency and cost savings, helping industries and organizations reduce electricity consumption and operational costs.

How GT Works:

1.Uses renewable resources instead of non-renewable resources: Green Technology promotes the use of natural energy sources that can be replenished continuously and do not harm the environment. **Ex:** Solar energy, wind energy, hydro power etc.

2.Improving energy efficiency: It encourages the development of devices and systems that use less energy while providing the same level of performance, helping to reduce electricity consumption. **Ex:** Energy-efficient appliances and LED lighting.

3.Reducing waste and harmful emissions: Green Technology aims to minimize pollution by reducing industrial waste and greenhouse gas emissions through cleaner production methods. **Ex:** Use of biodegradable materials and proper recycling.

4.Sustainable design and production: It promotes designing products, buildings, and systems that use fewer resources and have minimal environmental impact throughout their lifecycle. **Ex:** Green architecture, eco-friendly buildings, and sustainable manufacturing.

5.Promoting recycling and reuse of resources: GT encourages recycling materials and reusing resources to reduce waste and conserve natural resources. **Ex:** Recycling electronic waste and using recycled materials

6.Encouraging eco-friendly technologies: It supports the development and use of technologies that reduce environmental damage and improve sustainability. **Ex:** Electric vehicles, smart energy systems, and energy-efficient appliances.

Green IT's Social Impact:
1.Social Benefits: Green IT creates positive social change by improving quality of life, reducing pollution, and supporting sustainable development in society.
2.Better Public Health: Reduced emissions from energy-efficient technologies lead to cleaner air and fewer health problems such as respiratory diseases.
3.Employment Opportunities: Green IT creates new job opportunities in sectors such as renewable energy, recycling, and sustainable technology.
4.Digital Education Growth: Online learning platforms reduce the need for physical infrastructure and travel, making education more accessible and environmentally friendly.
5.Cost Savings: Organisations can reduce operational costs by saving energy, reducing paper usage, and improving efficiency.
Practical Example: Online meetings replace physical meetings, which reduces travel, saves fuel, lowers air pollution, and saves time, showing how Green IT benefits society in multiple ways.

Learning Organisations:
1.A learning organisation continuously improves by learning new ideas and adopting modern technologies.
2.It encourages employees to learn about sustainable practices and responsible use of IT resources.
3.These organisations promote innovation, environmental awareness, and efficient energy use.
4.They implement policies that support Green IT and sustainable development.
5.Ex: Training employees to switch off computers after office hours reduces electricity use, lowers costs, and increases hardware lifespan.

Green Social Stakeholders:
1.Definition: Green social stakeholders are individuals or groups that are affected by or involved in Green IT initiatives. They play an important role in promoting environmental sustainability.
2.Government: The government creates environmental policies, regulations, and incentives that encourage organisations to adopt green technologies.
3.Employees: Employees support Green IT by following eco-friendly practices in their daily work, such as saving energy and reducing paper usage.
4.Customers: Customers influence organisations by demanding environmentally friendly products and sustainable business practices.
5.IT Professionals: IT professionals design, develop, and maintain sustainable information systems that reduce energy consumption and environmental impact.
6.Society: The overall society benefits from Green IT through a cleaner environment, reduced pollution, and improved quality of life.
7.Ex: Government solar energy programmes encourage the use of renewable energy and reduce dependence on fossil fuels.
Role-Based View of Green IT:
1.Managers: Managers play a key role in promoting Green IT by creating environmental policies and sustainability strategies within the organisation.
2.They encourage the use of energy-efficient technologies and guide employees to follow environmentally responsible practices.
3.IT Departments: The IT department is responsible for implementing energy-efficient systems, managing green data centres, and using hardware that consumes less power.

4.They also maintain, upgrade, and monitor IT infrastructure to reduce energy consumption and electronic waste.
5.Employees: Employees support Green IT by following eco-friendly practices in their daily work, such as reducing paper usage and using digital communication.
5.Example: Switching off computers and other devices after use helps save electricity, reduce costs, and protect the environment.

Green User Practices:
1.Meaning: Green user practices are environmentally friendly habits followed by individual computer users to reduce energy consumption and environmental impact.
2.Low Screen Brightness: Reducing screen brightness lowers electricity consumption while still maintaining proper visibility.
3.Sleep Mode Activation: Using sleep mode during inactive periods helps conserve energy and prevents unnecessary power usage.
4.Cloud Storage Use: Using cloud storage reduces the need for physical storage devices, which lowers hardware production and electronic waste.
5.Laptop Preference: Using laptops instead of desktop computers saves electricity because laptops consume much less power.
6.Switching Off Devices: Turning off unused computers and electronic devices prevents standby power consumption and saves energy.
7.Ex: A laptop consumes much less electricity than a desktop computer, which helps reduce carbon emissions from electricity generation.

Attitude, Ethics, and Code of Conduct in Green IT:
1.Attitude and Subjectivity: Personal attitude plays an important role in adopting Green IT practices. When individuals value environmental protection, they are more likely to follow sustainable and responsible technology usage.
2.Positive attitudes towards sustainability encourage people to reduce waste, save energy, and support environmentally friendly IT practices.
3.Ex: Students using digital notes instead of paper notebooks help save paper, protect trees, and reduce waste. This shows how individual actions can create a positive collective impact.
4.Proper E-Waste Disposal: Old computers and electronic devices should be recycled through authorised centres to prevent toxic substances from polluting soil and water.
5.Honest Data Handling: Ethical IT practices include avoiding illegal software and ensuring responsible and secure management of digital data.
6.Energy Conservation: Responsible use of technology, such as switching off devices when not in use and using energy-efficient systems, helps reduce electricity consumption and carbon emissions.

Privacy and Security in Green IT:
1.Privacy and security are important in Green IT to protect digital information and systems from unauthorised access.
2.Organisations must ensure that data is stored, processed, and shared in a safe and secure manner.
3.Security measures such as strong passwords, encryption, firewalls, and access control help protect sensitive information.
4.These practices ensure that Green IT systems remain reliable while maintaining environmental efficiency.

5.Ex: Using encrypted cloud storage for organisational data prevents unauthorised access while reducing the need for physical storage devices.

Green Washing:

1.Green washing refers to the practice where companies make false or misleading claims about being environmentally friendly.
2.It is often used as a marketing strategy to attract customers who prefer eco-friendly products.
3.In reality, such companies may not follow genuine sustainable or green practices.
4.Green washing can mislead customers and reduce trust in real environmental initiatives.
5.Ex: A company advertising "eco-friendly products" while still using plastic packaging is misleading customers and continuing environmental harm.

Communications in Green Transformation:

1.Communication plays an important role in the success of green transformation within an organisation. It helps employees understand sustainability goals and green practices.
2.Policy Communication: Organisations should clearly explain their green policies, objectives, and expectations so that employees know how to follow them.
3.Training Sessions: Workshops and training programmes should be conducted to teach employees about energy conservation and sustainable practices.
4.Regular Updates: Organisations should share updates about progress and achievements in sustainability initiatives to keep employees informed and motivated.
5.Employee Engagement: Employees should be encouraged to give feedback and participate in green programmes. This increases commitment and helps achieve better environmental results.
6.Impact: Proper communication and training help employees follow green practices consistently, leading to reduced energy consumption and measurable environmental benefits.

Green HR Practices:

1.Integration: Green HR integrates environmental sustainability into HR activities such as recruitment, training, and daily operations to promote responsible organisational behaviour.
2.Paperless: Organisations use paperless offices, digital documentation, and electronic salary slips which reduce paper usage and help save trees and natural resources.
3.Remote Work: Work-from-home policies reduce employee commuting, which lowers fuel consumption, traffic congestion, and carbon emissions.
4.Structure: Organisations may change organisational structures by forming green teams or assigning roles for sustainability and environmental management.
5.Impact: Practices like virtual onboarding and sustainable office design reduce administrative costs, improve employee wellbeing, and show the organisation's commitment to environmental responsibility.

Green-Collar Workers:

1.Definition: Green-collar workers are professionals who work in jobs that support environmental sustainability and protection.
2.Work Areas: They are involved in fields such as renewable energy, recycling, waste management, and environmental conservation.
3.Skills: Green-collar workers require both technical skills and knowledge about environmental protection and sustainable practices.
4.Role: Their work helps reduce pollution, conserve natural resources, and promote environmentally friendly technologies.

5.Economic Impact: The demand for green-collar workers is increasing as more organisations adopt sustainable development practices.
6.Ex: Solar panel technicians generate clean energy from sunlight, helping reduce greenhouse gas emissions and supporting the green economy.

Green Virtual Communities:
1.Definition: Green virtual communities are online platforms where people share ideas, knowledge, and information related to environmental protection and sustainability.
2.Awareness: These communities help spread awareness about Green IT practices, climate change, recycling, and sustainable living.
3.Knowledge Sharing: Members can share resources, research, and experiences that help others adopt environmentally friendly practices.
4.Collaboration: People from different locations can work together on environmental projects without travelling, which saves energy and reduces carbon emissions.
5.Ex: Online tree plantation campaigns and environmental awareness drives allow people to participate digitally, reducing the need for physical travel.
6.Sustainable Future: Green virtual communities support a sustainable future by encouraging responsible technology use, collective action, and environmental protection.

Green Information Systems (GIS):
1.Definition: Green Information Systems refer to the design, development, implementation, and use of information systems in a way that reduces environmental impact and supports sustainability.
2.Purpose: GIS focuses on reducing energy consumption, carbon emissions, electronic waste, and excessive resource usage while improving organisational efficiency.
3.Benefits: It helps organisations save electricity, reduce paper usage, manage environmental data, and support eco-friendly decision making.
4.Ex: Using cloud storage instead of physical servers reduces power consumption and cooling needs. Companies like Google and Microsoft also use Green IS to optimise data centres and use renewable energy.

Green SDLC (Software Development Life Cycle):
1.Green SDLC is a modified version of the traditional SDLC that includes environmental goals.
2.Planning: Identify environmental objectives such as reducing power consumption.
3.Analysis: Measure carbon footprint and estimate resource requirements.
4.Design & Development: Create energy-efficient architecture and write optimized code that uses less CPU power.
5.Testing & Implementation: Test energy usage and deploy the system on green or energy-efficient servers.
6.Maintenance: Continuously monitor energy consumption and improve system efficiency.
7.Ex: A mobile app designed to use less background data saves battery and server energy.

Life Cycle Assessment (LCA):
1.Definition: Life Cycle Assessment (LCA) is a method used to measure the environmental impact of an IT product throughout its entire life cycle.
2.Manufacturing Stage: At this stage, LCA evaluates the environmental effects caused during the production of hardware and electronic components.

3.Usage Stage: It measures the energy consumption and environmental impact when the product is used by individuals or organisations.
4.Disposal Stage: LCA also considers the disposal or recycling of the product to reduce electronic waste and environmental pollution.
5.Sustainability: By analysing all these stages, LCA helps ensure sustainability from the creation (birth) of IT products to their final disposal.

Green Architecture Model:
1.Definition: The Green Architecture Model focuses on designing IT infrastructure in a way that reduces energy consumption and supports environmental sustainability.
2.Low Power Servers: It uses low power servers that consume less electricity while still maintaining good performance.
3.Efficient Networking: Efficient networking systems are designed to optimise data transfer and reduce unnecessary energy usage.
4.Virtual Machines: Virtualisation allows multiple virtual machines to run on a single physical server, reducing the need for extra hardware and saving energy.
5.Renewable Integration: Green architecture also supports the use of renewable energy sources such as solar or wind power to run IT systems.

Describing Green Information Systems (GIS):
1.Definition: Green Information Systems (GIS) refer to the design and use of information systems in a way that reduces environmental impact and promotes sustainable and responsible use of technology.
2.Components: GIS includes several elements such as **hardware, software, networks, organisational policies, and human behaviour** that work together to support environmentally friendly IT operations.
3.Objective: The main aim of GIS is to achieve **eco-efficient computing**, which means using technology efficiently to reduce energy consumption, minimise electronic waste, and conserve natural resources.
4.Applications: GIS supports important activities such as **environmental monitoring, smart grid systems, carbon accounting, and paperless office operations**, helping organisations track and reduce their environmental impact.
5.Benefits: By implementing GIS, organisations can improve operational efficiency, reduce electricity usage, lower carbon emissions, and support sustainable development.

GIS Requirements:
1.Overview: Green Information Systems (GIS) require proper technical, organisational, environmental, and economic support to successfully implement sustainable IT practices.
2.Technical Requirements: GIS requires energy-efficient hardware, virtualisation, cloud computing, and green data centres to reduce power consumption and improve system efficiency.
3.Organisational Requirements: Successful implementation of GIS needs management support, employee awareness, and clear green policies within the organisation.
4.Environmental Requirements: GIS should focus on low carbon emissions, proper waste recycling, and compliance with environmental standards such as RoHS to reduce environmental impact.
5.Economic Requirements: Green IT investments should provide cost savings and a good return on investment (ROI) while improving operational efficiency.

6.Ex: Replacing traditional desktop PCs with thin clients can reduce electricity consumption and lower maintenance costs for organisations.

Green Compliance:
1.Definition: Green compliance means following environmental laws, standards, and regulations while operating business activities.
2.Purpose: It ensures organizations reduce pollution and follow sustainable environmental practices.
3.Standards: Companies follow environmental standards such as ISO 14001 Environmental Management System.
4.Monitoring: These standards help organizations monitor pollution levels and track environmental performance.
5.Benefit: Green compliance helps companies minimize environmental damage and operate responsibly.

Green Audits and Types:
1.Meaning: Green audit is a systematic inspection of an organization’s environmental practices and impact.
2.Compliance Audit: This audit checks whether the company follows environmental laws and regulations.
3.Management Audit: It verifies whether the organization’s green policies are properly implemented.
4.Energy/Waste Audit: This audit measures energy consumption, waste generation, and carbon emissions.
5.Outcome: Green audits help organizations identify environmental problems and improve sustainability.

Emerging Carbon Issues:
1.Concept: Emerging carbon issues focus on managing and reducing greenhouse gas emissions.
2.Carbon Markets: Future systems may include carbon markets where companies buy and sell carbon credits.
3.Carbon Taxes: Governments may introduce higher carbon taxes to control pollution levels.
4.Monitoring: Organizations must track and report carbon emissions accurately.
5.Impact: These measures encourage industries to adopt cleaner and greener technologies.

Quantum Computing:
1.Definition: Quantum computing is an advanced computing technology based on principles of quantum mechanics.
2.Capability: It can solve complex problems faster than traditional computers.
3.Research: Quantum computing may help scientists discover new materials and environmental solutions.
4.Carbon Capture: It could help design technologies that remove CO_2 from the atmosphere.
5.Environmental Impact: This technology may improve climate modelling and environmental analysis.

Collaborative Environmental Intelligence:

1.Meaning: Collaborative environmental intelligence refers to sharing environmental data among organizations and governments.

2.Technology: It uses tools such as **AI, big data, and analytics** to study environmental changes.

3.Collaboration: Different organizations work together to predict environmental risks.

4.Decision Making: Shared data helps improve planning and sustainable development decisions.

5.Benefit: It strengthens global cooperation in solving environmental problems.

Technology Environmental Benefit:

1.Concept: Technology can provide environmental benefits when designed and used responsibly.

2.Eco-Design: Products are designed to be easily recyclable and environmentally friendly.

3.Renewable Energy: Technologies like solar and wind power reduce dependence on fossil fuels.

4.Innovation: New technologies help develop clean fuels such as hydrogen.

5.Sustainability: These benefits help reduce pollution, conserve resources, and support sustainable development.